Summary

Immunology began as a study of the response of the whole animal to infection. Over the years, it has become progressively more basic, passing through phases of emphasis on serology, cellular immunology, molecular immunology and immunogenetics.

The thymus and bone marrow are the primary lymphoid organs. The primary lymphoid organs provide sites for the development and maturation of B and T lymphocytes.

The secondary lymphoid organs function to capture antigen and provide sites where lymphocytes interact with that antigen and undergo clonal proliferation and differentiation into effector cells. The lymphatic system drains the tissue spaces and interconnects many organized lymphoid tissues. The spleen, lymphnodes and mucosal associated tissues (GALT and SALT) are secondary lymphoid organs. Lymph nodes are specialized to trap antigen from regional tissue spaces, whereas the spleen traps blood- borne antigens.

The cells that participate in the immune response are white blood cells or leukocytes. All of the white blood cells develop from a common pluripotent stem cell in hematopoiesis. Lymphocytes are the central cells of the immune system and are responsible for acquired immunity. The other types of white blood cells play ancillary roles such as engulfing and destroying microorganisms, presenting antigens and secreting cytokines. Basophills and mast cells are non phagocytic granulocytes that play a role in allergic responses. Eosinophils are motile phag ocytic cells. Their phagocytic role is less important than that of neutrophils. They play a role in the defense against parasitic organisms. Macrophages and neutrophils are the accessory cells of the immune system that phagocytose and degrade antigens. Dendritic cells are antigen presenting cells. They play an important role in T_H cell activation by processing and presenting antigen bound to class II MHC molecules. Lymphocytes can be subdivided into B lymphocytes, T lymphocytes and null cells (NK cells). The two major subpopulations of T lymphocytes are T helper (T_H) cells and T cytotoxic (T_C) cells.

Antibodies are a group of glycoproteins present in the blood tissue fluids and mucous membranes of vertebrates. All immunoglobulins have a basic structure composed of four polypeptide chains (two lights and two heavy) connected to each other by disulphide bonds. In any given antibody molecule, the constant region contains one of five basic heavy chain sequences (γ, α, μ, δ and ε) and one of two basic light chain sequences (k or γ). There are three major effector functions of immunoglobulins are Opsonization, Complement activation and Antibody-dependent cell-mediated cytotoxicity (ADCC).

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